Assignment 2: Topic Exploration and Initial Direction

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Part 1: Topic Ideas and Descriptions

This section describes three potential topics for a literature review, each rooted in different aspects of technology and education.

Topic 1: Vision-Language Models for Construction Site Monitoring

Construction sites are complex and constantly changing. This topic focuses on how vision-language models (VLMs) can understand what happens across space and time at construction sites. The key interest is moving beyond simple detection of objects like excavators or hardhats. Instead, the focus is on understanding sequences of events and spatial relationships over time. VLMs represent an intersection between AI and construction management that could improve how we monitor safety and progress.

Topic 2: Large Language Models for Assessing Inclusiveness in Educational Materials

Educational materials should represent everyone fairly. This topic explores how Large Language Models (LLMs) can help assess whether textbooks and educational documents include diverse perspectives and avoid biases. The emergence of LLMs makes it possible to automate this assessment. However, much work remains to understand how to prompt and train these models effectively, what benchmarks to use, and how to properly annotate data for accuracy.

Topic 3: Machine Learning in Construction Workforce Management

Construction projects depend on managing workers well. This topic examines how machine learning is being used in construction to manage teams. It includes predictive modeling and data analytics for worker safety, scheduling decisions, and how to allocate resources effectively. The field is growing, but there is still much to discover about what works and what needs improvement.

Part 2: Scope and Feasibility Analysis

Topic 1: Vision-Language Models for Construction Site Monitoring

Scope: This topic focuses on one specific technology (VLMs) in one specific place (construction sites) for specific goals (safety and progress tracking). It narrows further because the focus is on understanding what happens over time and space, not just spotting objects.

Feasibility: Papers on this exist and more are coming out. But finding the right ones is hard. When I search for computer vision in construction, I get thousands of papers about basic deep learning or simple object spotting. To make this work, I need to search carefully and only include papers that actually use vision-language models. This is doable but needs precision from the start.

Why it works for six weeks: The focus is narrow enough that I can manage it in six weeks. The real work is not finding papers - it is filtering them down to the ones that matter.

Topic 2: Large Language Models for Assessing Inclusiveness in Educational Materials

Scope: This is about using LLMs to answer a specific question: do educational materials treat everyone fairly and include different viewpoints? I need to understand both how to use the models and how to measure if they work.

Feasibility: Papers exist on LLMs. Papers exist on whether education is fair and inclusive. But papers specifically about using LLMs to check if materials are inclusive are not common yet. That said, papers on AI fairness, finding bias, and educational equality all add pieces to this puzzle.

Why it works for six weeks: The scope is manageable. I can pull ideas from several research areas. That said, it is broader than Topic 1, so I might spend more time just organizing the papers than going deep into analysis.

Topic 3: Machine Learning in Construction Workforce Management

Scope: This is the widest of the three. It includes several different uses: keeping workers safe, scheduling, and deciding how to use resources. It could involve many types of machine learning methods.

Feasibility: Papers on machine learning in construction exist. Papers on workforce management exist. But the intersection is very broad. Papers on deep learning or construction could easily not be relevant.

Why it is less suitable for six weeks: The topic is too wide. I could spend all six weeks just figuring out what counts and what does not. It needs to be much more narrow to work.

Part 3: Initial Thinking About Review Methodologies

For each topic, different review approaches might work better or worse.

Topic 1 Methodology

A scoping review makes sense for this topic. The goal is to see what exists in terms of how VLMs are used in construction monitoring right now. I need to know what is out there, what doesn't work, and where things are going. A scoping review shows the range of work and finds holes that future research could fill. Another option is a systematic review if I narrow the question to comparing how different VLM methods perform on specific tasks.

Topic 2 Methodology

An integrative review fits this topic. The reason is simple: understanding how to assess whether materials are inclusive requires bringing several things together. I need technical knowledge about how LLMs work, knowledge about what makes education fair and inclusive, and knowledge about how to spot bias. These come from different research areas. An integrative review is designed to do exactly this - pull together different kinds of knowledge to understand something better.

Topic 3 Methodology

A narrative review works best here. This topic is wide, and a narrative review is better for that. The goal is to see what machine learning is doing in construction workforce settings. Narrative reviews look at patterns and themes without needing rigid, formal search rules. That is what this topic needs.

Part 4: Selected Topic and Rationale

Selected Topic: Vision-Language Models for Construction Site Monitoring

I choose Topic 1 for several reasons.

First, it connects directly to my PhD work. I am learning about spatiotemporal reasoning and vision-language models right now. Doing this literature review will give me a solid base of knowledge about what already exists, and I will find the papers I need to understand going forward.

Second, the scope is right. It is narrow enough that the search stays manageable, but wide enough that there is plenty to read. The other two options do not connect to my work (Topic 2) or they are just too big for six weeks (Topic 3).

Third, it matters in the real world. Construction companies are already using monitoring technology. If VLMs can make this better, that has real value for safety.

Fourth, this prepares me for publishing. As people pointed out in the discussion, the move from just identifying objects to understanding what happens over time and in space is the interesting frontier. That is where good publications come from.

Part 5: Keywords for Selected Topic

The following search strings will guide my literature search for Topic 1:

1. ("vision-language model" OR "vision language model" OR VLM) AND construction AND (monitoring OR detection OR tracking)
2. ("vision-language" OR multimodal) AND construction AND ("safety" OR "hazard" OR "worker protection")
3. "spatiotemporal reasoning" AND construction AND (monitoring OR tracking OR prediction)

4. ("multimodal learning" OR "vision-language grounding") AND construction AND (site OR project)
5. (VLM OR "foundation model") AND ("construction site" OR "building site") AND ("real-time" OR "automated")

These search strings focus on finding papers that actually use vision-language models or multimodal approaches in construction. They avoid general deep learning or basic computer vision papers. They also exclude generic construction or AI papers that do not address the vision-language intersection.

Part 6: Reflection

This assignment helped me think clearly about what I actually want to study. I started with three ideas, and by thinking about what each one needs and what fits my work, Topic 1 became obvious.

The biggest problem I will have is search precision. When you search for computer vision and construction, you get so many papers about basic deep learning and simple object detection. I cannot use all of those. I need rules that say papers must actually use vision-language models, not just any computer vision. That is harder work than a simple search, but I have to do it.

Colleagues in the discussion made a good point: I should focus even more narrowly. Several people said to pick either safety or progress monitoring instead of both. That is smart. When I start searching, I will focus mainly on safety monitoring.

I also noticed something Luis said: I should find papers that show real projects, not just research in labs. A model can look great in papers but if nobody actually uses it on real construction sites, that is a real limitation. So I need another rule: papers should show either real cases or serious barriers to real-world use.

I feel good about this choice. It is specific but not too narrow, it aligns with what I am already doing, and it matters. Next I need to write down a clear research question and make my search strategy really tight.